DEAN HANSON

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PROFESSIONAL SUMMARY

Specializes in advanced HVAC engineering for complex facilities, encompassing central plant design, cleanrooms, process exhaust systems, and high-reliability environments. Brings strong expertise in controls design, sequences of operation, and construction-phase support, paired with clear technical communication and client leadership. Experienced in standardization, workflow automation, and AI integration within engineering practices, leveraging BIM and data-driven tools to enhance project delivery and reduce design risk.

HVAC EXPERIENCE IN THE AEC INDUSTRY

BRPH, *HVAC Project Engineer* **July 2019 – Present,** *Space Coast & Orlando, FL*

BRPH Role Scope & Project Delivery

- ➢ Manages full HVAC design lifecycle including system selection, calculations, drafting, site visits, specifications, design narratives, cost estimating, and client presentations.
- ➢ Serves as the primary technical interface for clients, leading meetings, resolving complex design challenges, and answering high-stakes questions in real time.
- ➢ Designs specialized environments including ISO-class cleanrooms, SCIFs, server rooms, and equipment-intensive facilities with evolving requirements.
- ➢ Engineers and coordinates advanced systems including chilled water and condenser water plants, cooling towers, boilers, DOAS, VAV AHUs, and complex process exhaust systems with scrubbers.
- ➢ Supports projects with construction budgets ranging from $100k to $100M.
- ➢ Key clients include SpaceX, Lockheed Martin, Boeing, Blue Origin, L3Harris, Northrop Grumman, NASA, Cape Canaveral SFS, and Patrick SFB.

BRPH Technical Leadership, Innovation, Culture

- ➢ Nationwide HVAC AI Subject Matter Expert and AI Committee Lead, driving AI integration across engineering workflows.
- ➢ Develops enterprise-level AI tools using GPT-5.2 and frontier LLMs, RAG architecture, embedded Python, and curated engineering documentation.
- ➢ Modernizes internal standards of operation across all project phases including calculations, selections, drawings, schedules, controls, and specifications.
- ➢ Overhauled and standardized 15+ years of legacy controls diagrams, sequences of operation, and schedules into a unified national library.
- ➢ Proactively develop departmental templates and documentation to improve efficiency, consistency, and design quality.
- ➢ Active leader in BRPH's nationwide wellness committee, organizing events and authoring internal communications

Surna, *HVAC Engineer Intern* **Summer 2018,** *Boulder, CO*

- ➢ Designed and drafted process cooling systems for controlled-environment agriculture facilities
- ➢ Performed load calculations and selected FCUs, chillers, pumps, air handlers, dehumidifiers, CO_2 exhaust systems, and condensate equipment
- ➢ Collaborated with engineers, project managers, sales teams, clients, and third-party manufacturers

WEEKLY SOFTWARE STACK

Highly competent in leveraging Bluebeam, Revit with critical plug-ins (Ideate, Dynamo and PyRevit), GPT with markdown files, embedded Python & trained agents, Trace 700+3D exports, Microsoft Excel and Word templates

EDUCATION & CERTIFICATIONS

The Pennsylvania State University, Bachelor of Science, Mechanical Engineering, May 2019 Graduating Class
Engineer In Training (May 2019) + LEED Green Associate (January 2019)